Student Worksheet

Name: _____ Period: _____ Date: _____

This guide will help you transform your single-player bottom-paddle starter code into a two-player side-to-side paddle game on an 800 × 600 canvas. Instead of rewriting your code from scratch, you will modify your existing decisions step-by-step.

1. Variable Adjustments (TODO 0)

Your original code tracked a horizontal paddle on the bottom wall using paddleX. Because the new paddles sit on the left and right sides, they must move vertically up and down.

- ☐ Task: Rename paddleX to paddle1Y (for Player 1) and create a second variable paddle2Y (for Player 2).
- ☐ Task: Create two separate hit-detection tracking states: over1 and over2 (instead of just one over).
- ☐ Task: Initialize your ball's x and y starting variables to the exact center of your new screen.

2. Canvas & Wall Adjustments (TODO 2)

Update createCanvas(400, 400) to createCanvas(800, 600). Because the canvas size changed, all of your wall coordinates must change too.

- ☐ Top & Bottom Walls: The ball must bounce when it hits the top wall (y < 10) or the bottom wall. What is the bottom wall coordinate now? Hint: Look at your new canvas height.
- ☐ Misses: In the starter code, a miss happened at the bottom (y > 370). In a side-to-side game, a miss happens when the ball escapes past the left wall (x < 0) or past the right wall. Update your conditional check to look for x boundaries instead of y boundaries.

3. Distinct Player Controls (TODO 3)

Instead of one paddle following the mouse horizontally, you must program two separate inputs:

- ☐ Player 1 (Left Paddle): Use if (keyIsPressed) to check if the user is pressing a specific key (like 'w' to subtract from paddle1Y and 's' to add to paddle1Y).
- ☐ Player 2 (Right Paddle): Assign its vertical position directly to mouseY.
- ☐ The Constraint: Use the constrain() function to lock both paddles within the screen limits (0 to 520).

4. Flipped Over Checks (TODO 4)

Your starter code checked if the ball was horizontally within the paddle's width (x > paddleX && x < paddleX + 80). For side paddles, you must flip this math to check the vertical axis.

- ☐ Task: Change the logic so over1 becomes true when the ball's y position is greater than paddle1Y AND less than paddle1Y + 80. Duplicate this logic for over2 using paddle2Y.

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5. Side Paddle Collisions (TODO 5)

Your paddles are now anchored on the left and right walls.

- ☐ Left Paddle Hit: Check if the ball's horizontal position (x) has traveled all the way to the left side AND over1 is true. If both are true, reverse your horizontal velocity (speedX = -speedX).
- ☐ Right Paddle Hit: Check if the ball's horizontal position (x) has traveled all the way to the right edge of the screen AND over2 is true. If both are true, reverse your horizontal velocity.

6. Rendering Your Objects (TODO 6)

In your drawing block, switch your rect() orientation. Your paddles should be narrow horizontally but long vertically.

- ☐ Player 1 Rectangle: Draw it at a fixed left x coordinate (like 7), using paddle1Y for its vertical position. Give it a width of 14 and a height of 80.
- ☐ Player 2 Rectangle: Draw it at a fixed right x coordinate (like 777), using paddle2Y for its vertical position. Give it a width of 14 and a height of 80.